

Basic Principles of Medicine 1

Module : Foundations to Medicine

DLA Title: Prokaryotic Gene Regulation

To be viewed before

Eukaryotic Gene Regulation Lecture

Name of Lecturer: Dr. Mary Maj
St Georges University ©



DLA: Prokaryotic Gene Regulation

Dr. Mary Maj email: mmaj@sgu.edu

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Required Reading for Lecture and DLA

Lippincott Illustrated Reviews in Biochemistry, 8th Edition, Chapter 33

Recommended Reading for Lecture and DLA

Korf & Irons 4th edition:

Figure 2.3– Mutation within a promoter– p 23,

Korf Hot Topic 1.1 – RNAi p12 - 13;

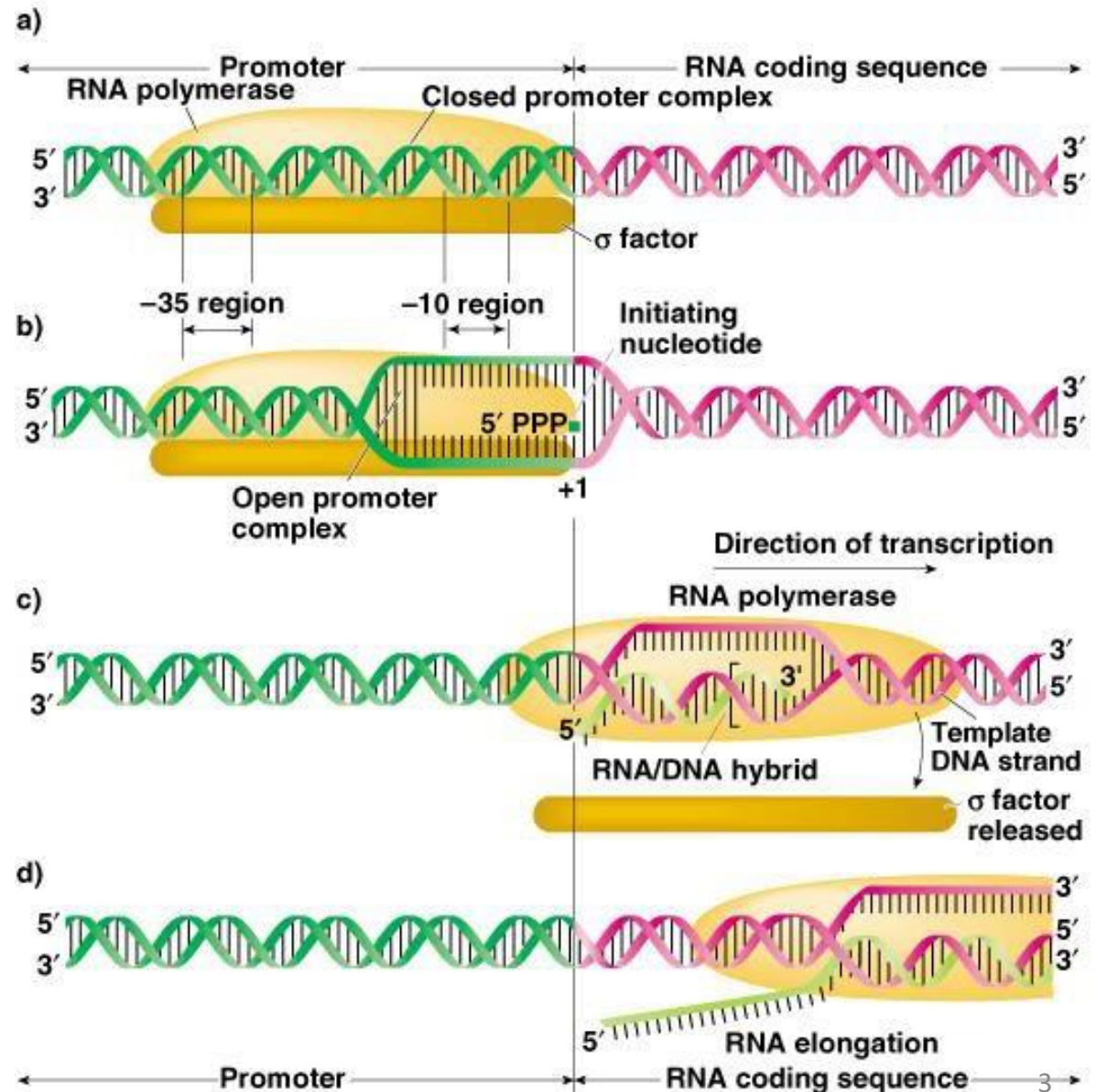
Korf p 8-11

Objectives Prokaryotic Regulation

SOM.MK.III.BPM1.1.FTM.3.GNET.0090	Differentiate between constitutive, inducible and repressible gene regulation
SOM.MK.III.BPM1.1.FTM.3.GNET.0091	Analyze the common component parts of general “operon” systems
SOM.MK.III.BPM1.1.FTM.3.GNET.0092	Identify the specific conditions by which individual operons will be transcribed: The influence of regulatory proteins which can activate or repress expression
SOM.MK.III.BPM1.1.FTM.3.GNET.0093	Identify the specific conditions by which individual operons will be transcribed: Conditions which influence positive and negative regulation for each particular operon
SOM.MK.III.BPM1.1.FTM.3.GNET.0094	Explain the significance of mutations within an operon and how they can affect gene regulation
SOM.MK.III.BPM1.1.FTM.3.GNET.0095	Compare & contrast eukaryotic & prokaryotic gene regulation (make your own chart)

General Transcription in Bacteria

General transcription in bacteria occurs by binding of RNA polymerase to the promoter region of a gene, the subsequent unwinding of the DNA double helix to reveal a single stranded DNA **template**, and the synthesis of an RNA transcript.



How is genetic expression regulated in prokaryotes?

- Bacteria regulate gene expression in response to environmental conditions
- *E. coli* has about 4000 different possible polypeptide chains encoded for on the genome
- Some proteins have only 5-10 molecules in the cell
- Others, such as ribosomal proteins and proteins in the glycolytic pathway can have as many as 100,000 copies in the cell
- **Some proteins have very few copies (basal levels) which can be increased dramatically when required**
- There are regulatory mechanism to control the expression of genetic information

Majority of Prokaryotic Gene Regulation takes place at the Transcriptional Level

- Can also be regulated at translation
- Regulation of protein half-life (stability of protein in cytosol)

Constitutive vs. Inducible Genes

- Bacteria have efficient mechanisms to turn transcription of genes off and on in response to environment and cellular activity
 - Nutrients available in environment?
 - Specific nutrients require specific degradative enzymes
 - Cellular activity including: replication, cell division, protein synthesis
 - Are there enough nucleotides in the cell for replication?
 - Are there enough amino acids in the cell for protein synthesis?

Constitutive vs. Inducible Genes

Inducible Genes

- Enzymes which are synthesized only when required based upon the chemical makeup of the environment are called **Inducible enzymes** from **Inducible genes** which are **Inducibly Expressed**
 - Turning off inducible genes referred to as ***repression*** or ***repressible***
 - The presence of a specific molecule inhibits gene expression

Constitutive Genes

- Enzymes which are produced continuously, regardless of the chemical makeup of the environment are called **Constitutive Enzymes** from **Constitutive Genes** which are **Constitutively Expressed**
 - Typically, low level, constant expression
 - Some gene products of constitutive genes regulate the expression of inducible genes

Inducible Gene Regulation:

Either positive or negative

A gene (or group of genes) can have both positive and negative control as seen with the metabolism of lactose in *E. coli*

- **Negative control**

- Genetic expression occurs unless it is shut off by some sort of a regulator molecule

- **Positive control**

- Genetic expression occurs only if a regulator molecule directly stimulates RNA production

Prokaryotic Cells Have 3 Types of Regulator Molecules Affecting Gene Expression

- **Repressors**

- **Activators**

- **Inducers**

} Proteins produced in the cell which bind to DNA regions close to the genes that they will control

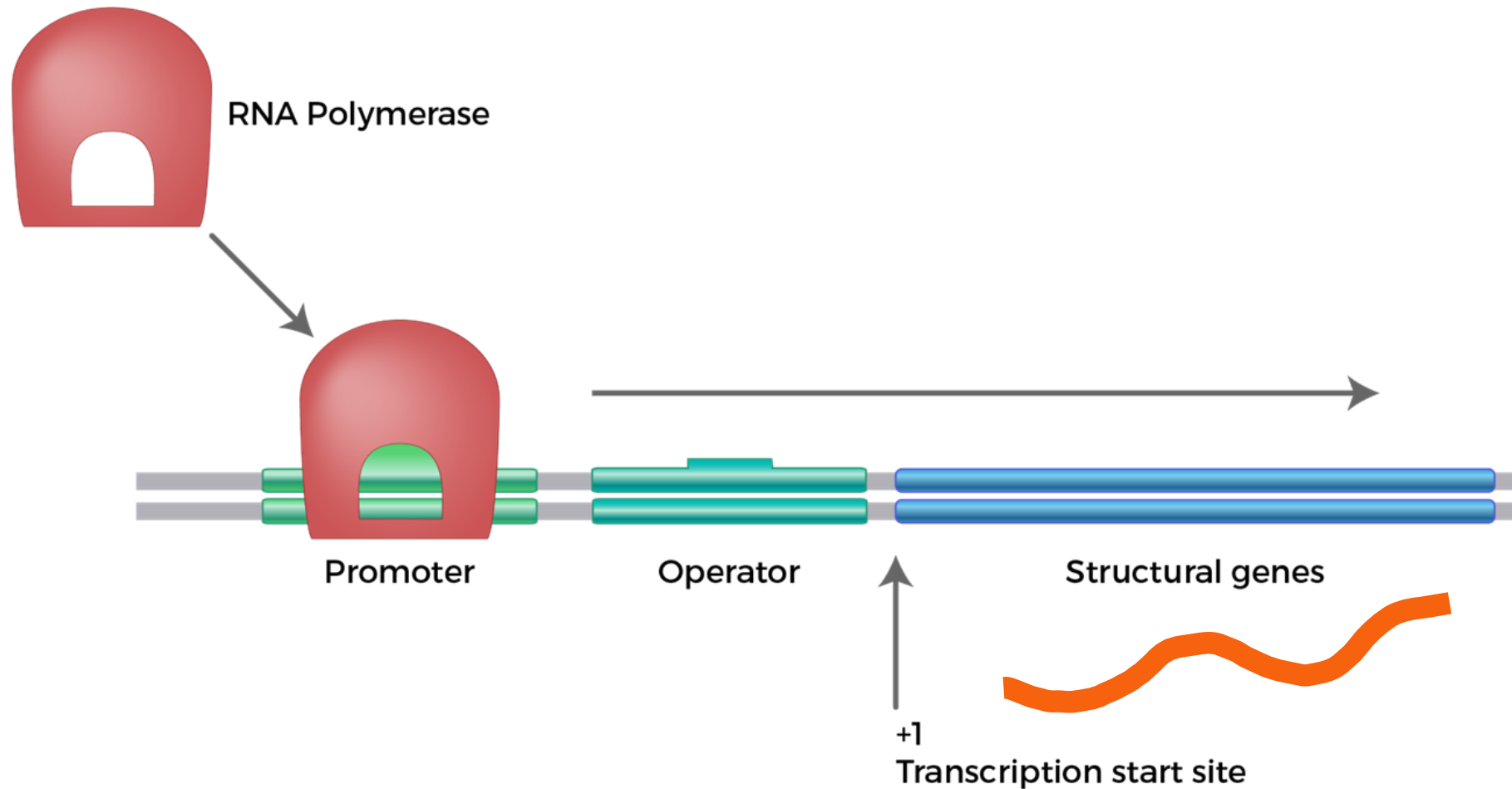
} Small molecules, either produced in the cell or taken up in to the cell, which can either activate or repress transcription, depending on needs of the cell at a particular time

Regulatory DNA Regions in Prokaryotes

- Structural Genes encoding for enzymes are often organized in clusters (polycistronic) **depending on function or pathway**
- Transcription of these genes is under the control of a single regulatory DNA region located upstream of the cluster that they control
- **Regulatory regions** are DNA sequences on the same strand as the polycistronic cluster (**cis-acting** or **cis-elements**)
- **Trans-acting** elements are the protein molecules that bind the cis-acting sites
- Trans-acting elements can positively or negatively regulate the transcription of the structural genes
 - Negatively by turning off transcription
 - Positively by turning on transcription

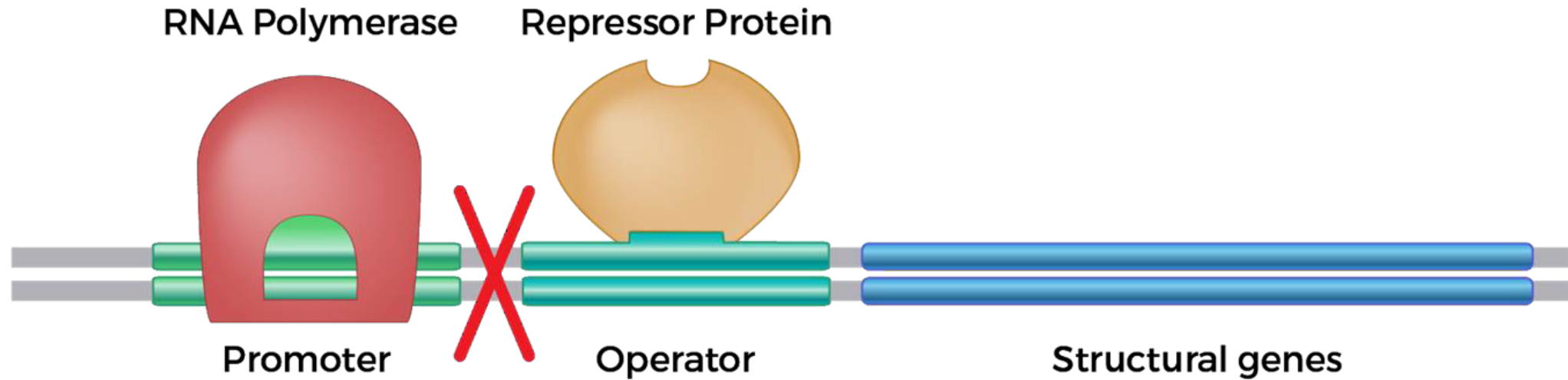
Regulatory Regions: Promoters and Operators

What is a Promoter?



- RNA polymerase recognizes the promoter sequence (Pribnow box) and will bind to it
- RNA polymerase will scan along the DNA till it finds the transcription initiation (start) site for the structural gene
- RNA polymerase will begin transcribing the RNA of the structural genes

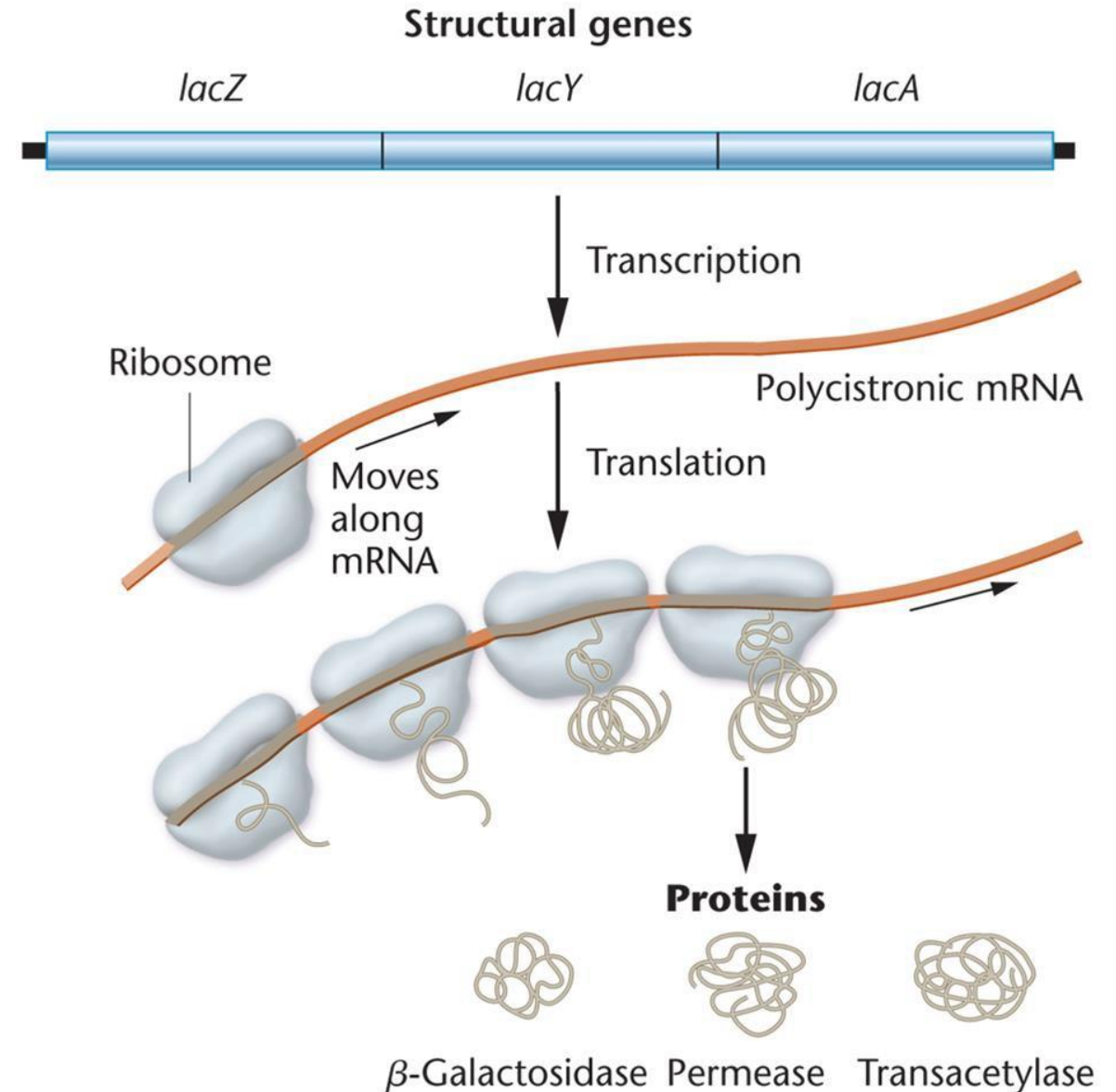
What is an operator?



- The operator is a sequence of DNA downstream of the promoter which is recognized by a repressor protein
- If the repressor protein is bound, RNA polymerase cannot synthesize RNA

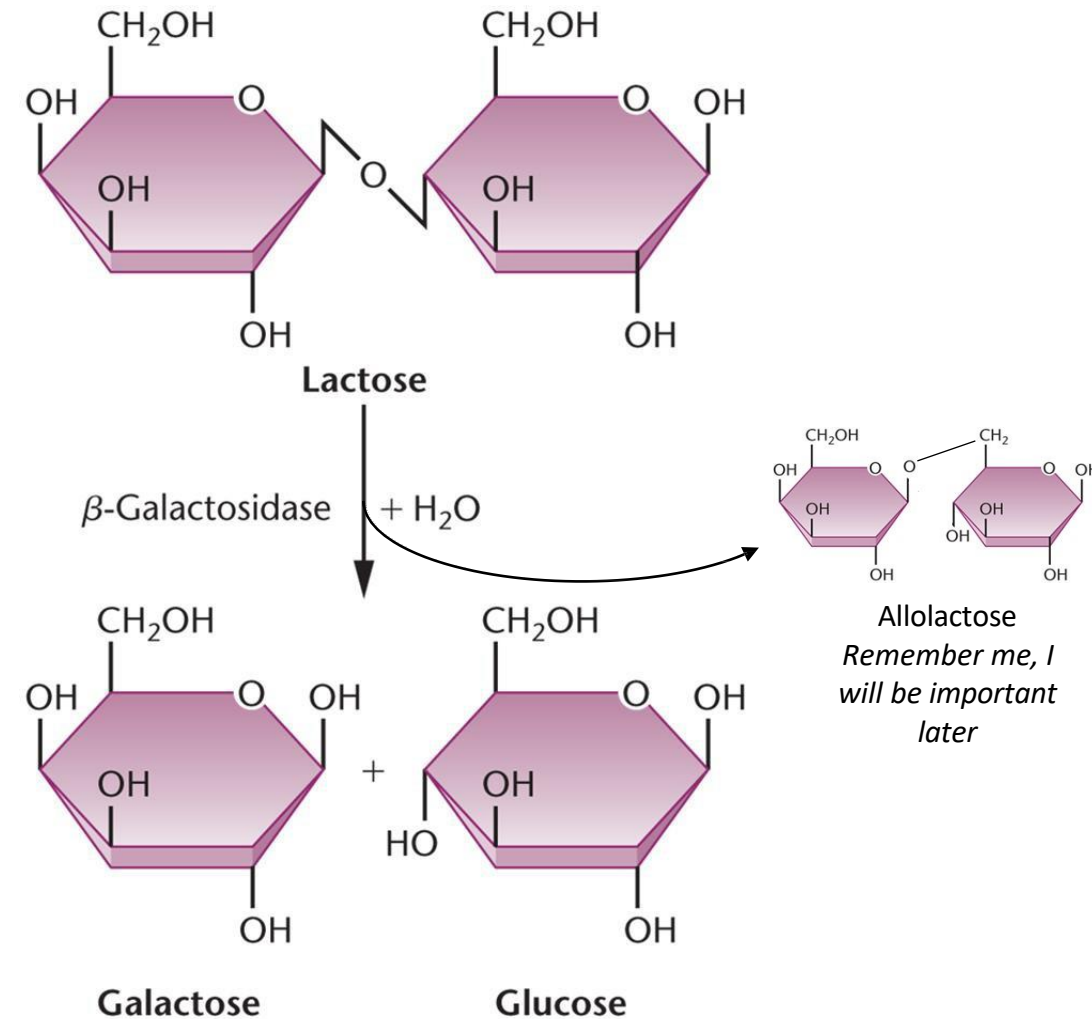
Transcription of the Structural genes

- RNA polymerase will bind to the promotor and transcribe the structural genes in a single unit (polycistronic RNA) including:
 - *lacZ*, *lacY* and *lacA*
- The single RNA is translated into three gene products
- Entire gene cluster functions to provide rapid response to the presence or absence of lactose



Catabolic conversion of the disaccharide lactose

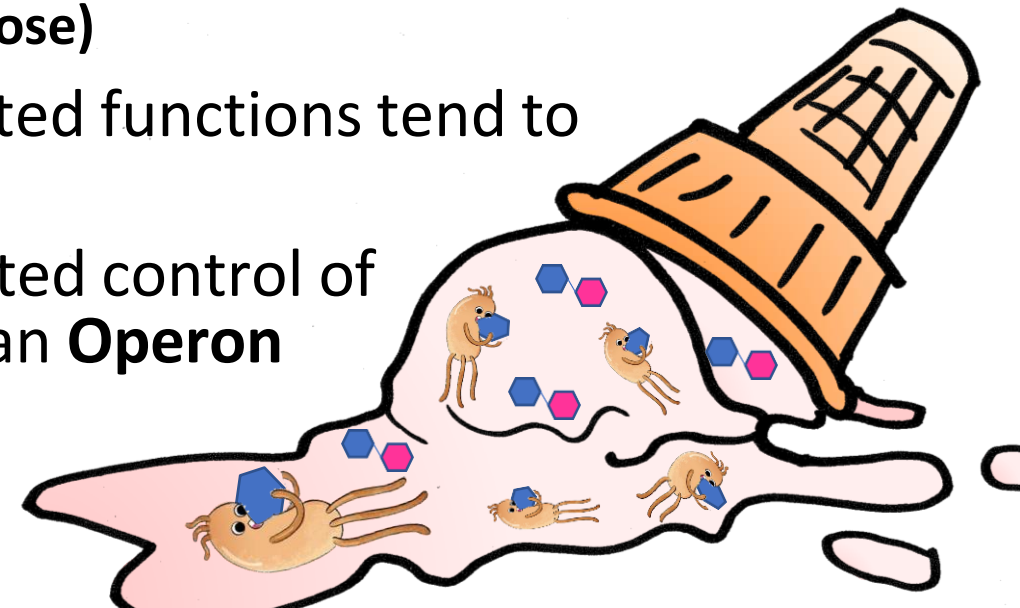
- Lactose can be metabolized for energy by bacteria
- First step is the cleavage of lactose to galactose and glucose
- The enzyme is **β -galactosidase**
 - A minor activity of β -galactosidase is the conversion of lactose to **allolactose**
- Glucose can be metabolized directly but galactose must be converted glucose-6-phosphate before it is metabolized by the bacteria for energy



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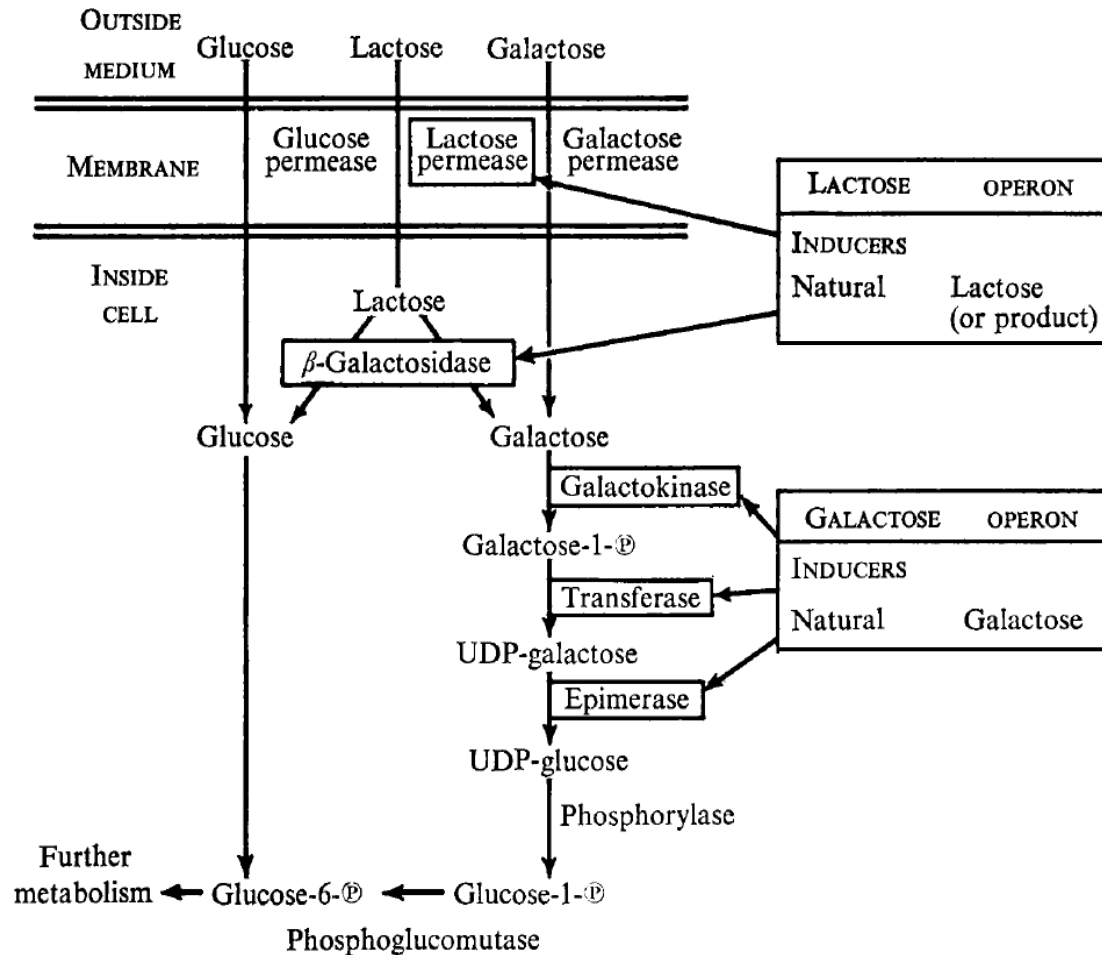
Lactose Metabolism in *E. coli* is regulated by an Inducible System

- Francois Jacob and Jacques Monod were studying metabolism in *E. coli*
- Glucose is the preferred food source for bacteria
- If glucose becomes depleted in the environment and lactose is present,
 - The enzymes responsible for the metabolism of lactose increases rapidly from a few molecules to thousands per cell
 - The enzymes responsible for lactose metabolism are inducible
 - **Lactose is the inducer (after it is converted to allolactose)**
- In bacteria, genes that encode enzymes with related functions tend to be organized in clusters
- These gene clusters are often under the coordinated control of a single regulatory region, entire region is called an **Operon**



Glucose is the Preferred Sugar Source for Bacteria

No need to memorize this diagram



- Glucose is metabolized most efficiently by bacteria, first step is conversion to glucose-6-P
- If there is no glucose in the environment but there is lactose, an enzyme called **β -galactosidase** will cleave lactose to glucose and galactose
- The glucose portion is metabolized easily to glucose-6-P
- The galactose must go through many steps to be converted to glucose-6-P
- Metabolism of lactose is not as efficient as pure glucose

The Lactose (*lac*) operon

Jacob and Monod studied the regulation of genes required for the metabolism of lactose in bacteria (*coordinate regulation*)

Mutation analysis to identify genes regulating lactose metabolism



François Jacob
(1920 - 2013)

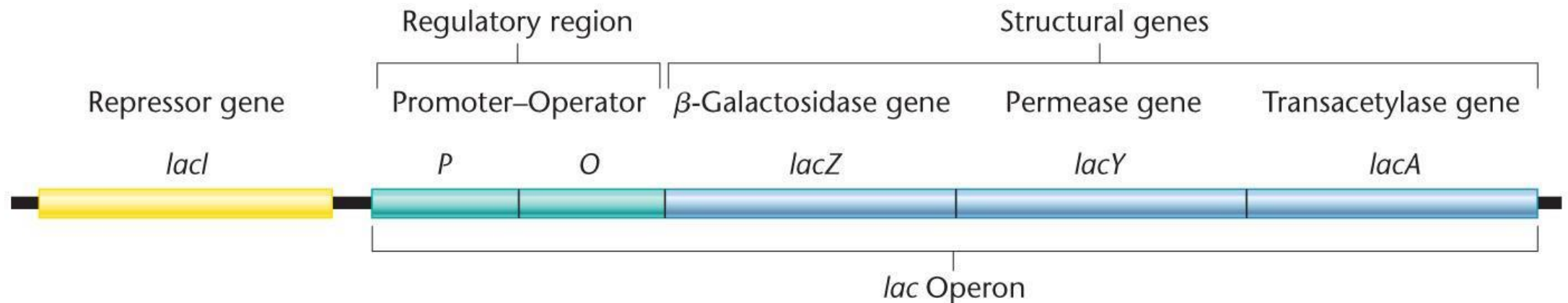


Jacques Monod
(1910 – 1976)

Won the Nobel Prize in Physiology or Medicine, 1965

The *lac* Operon

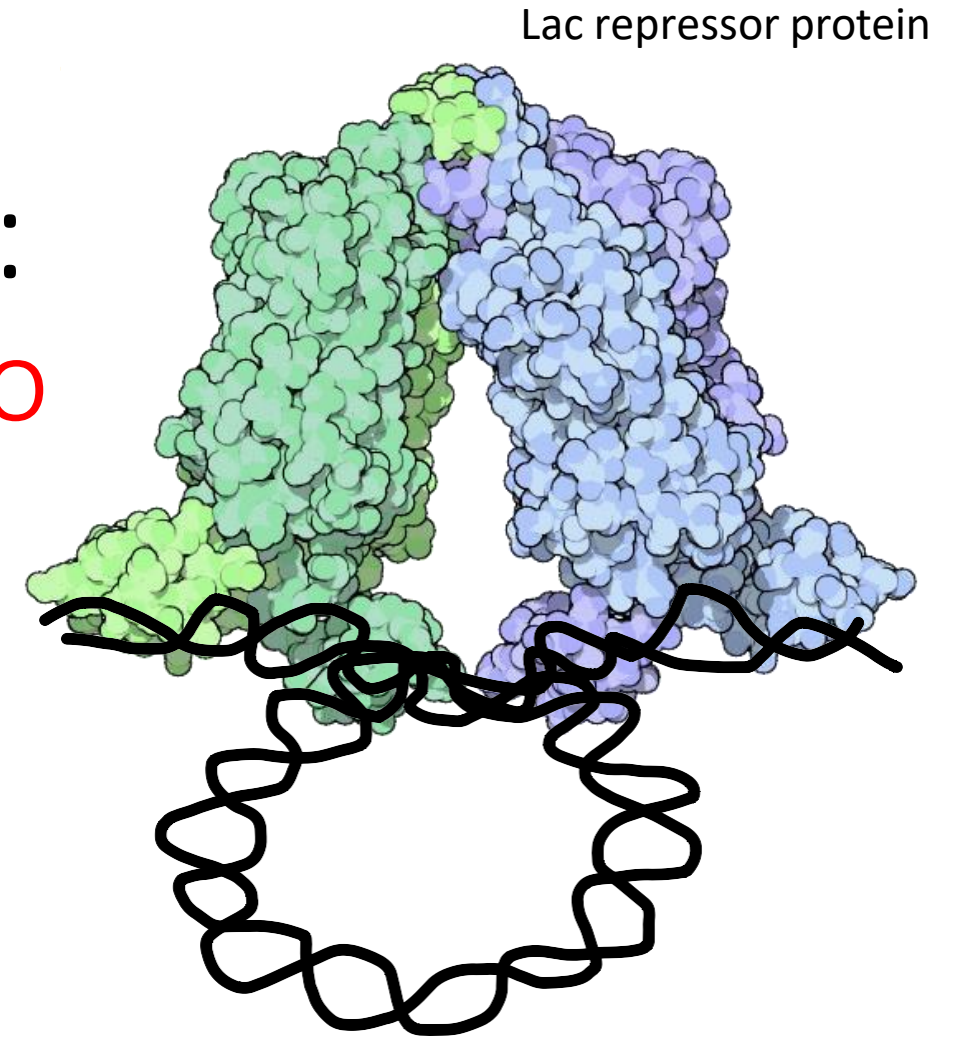
- There are 3 enzymes required to metabolize lactose
 - β -galactose, permease and transacetylase
- The 3 genes are called **Structural Genes** and they are clustered together in a tip to tail fashion (**polycistronic**)
 - *lacZ*, *lacY* and *lacA*
- One **regulatory region, promoter and operator**
- One **repressor gene, *lacI*** which encodes for a ***repressor protein***



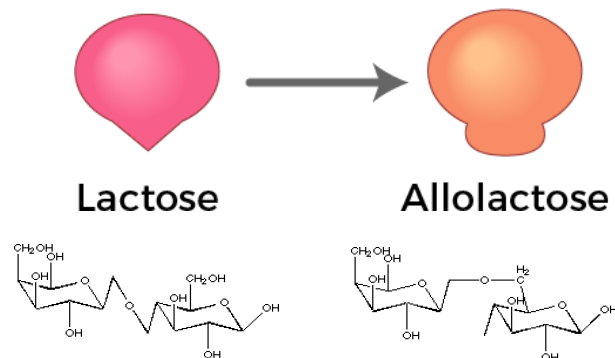
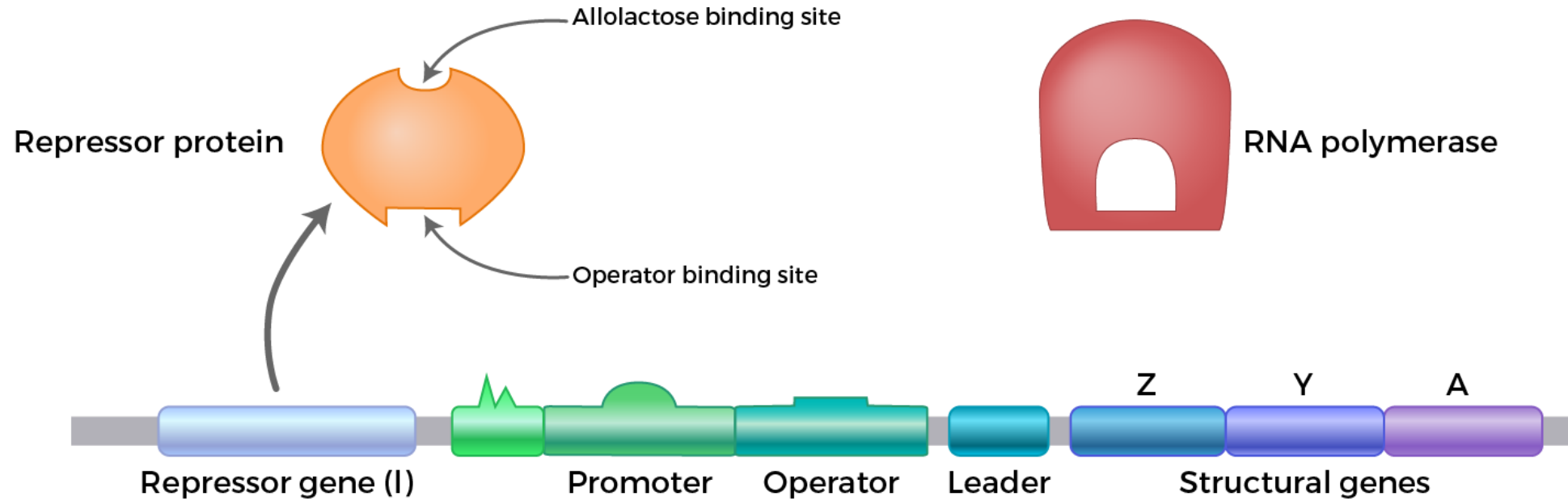
Proteins of the lac operon

- *lacZ*
 - Encodes for β -galactosidase, an enzyme that converts the disaccharide lactose to monosaccharides glucose and galactose
- *lacY*
 - Encodes for permease, an enzyme that facilitates the entry of lactose into the bacterial cell
- *lacA*
 - Encodes for transacetylase which may be involved in the removal of toxic by-products of lactose digestion from the cell
- *lacI* 
 - Encodes for a repressor protein that regulates the transcription of *lacZ*, *lacY* & *lacA*

Control of the lac operon:
No Lactose around so you do
not need the structural
genes synthesized



Introducing all of the participants

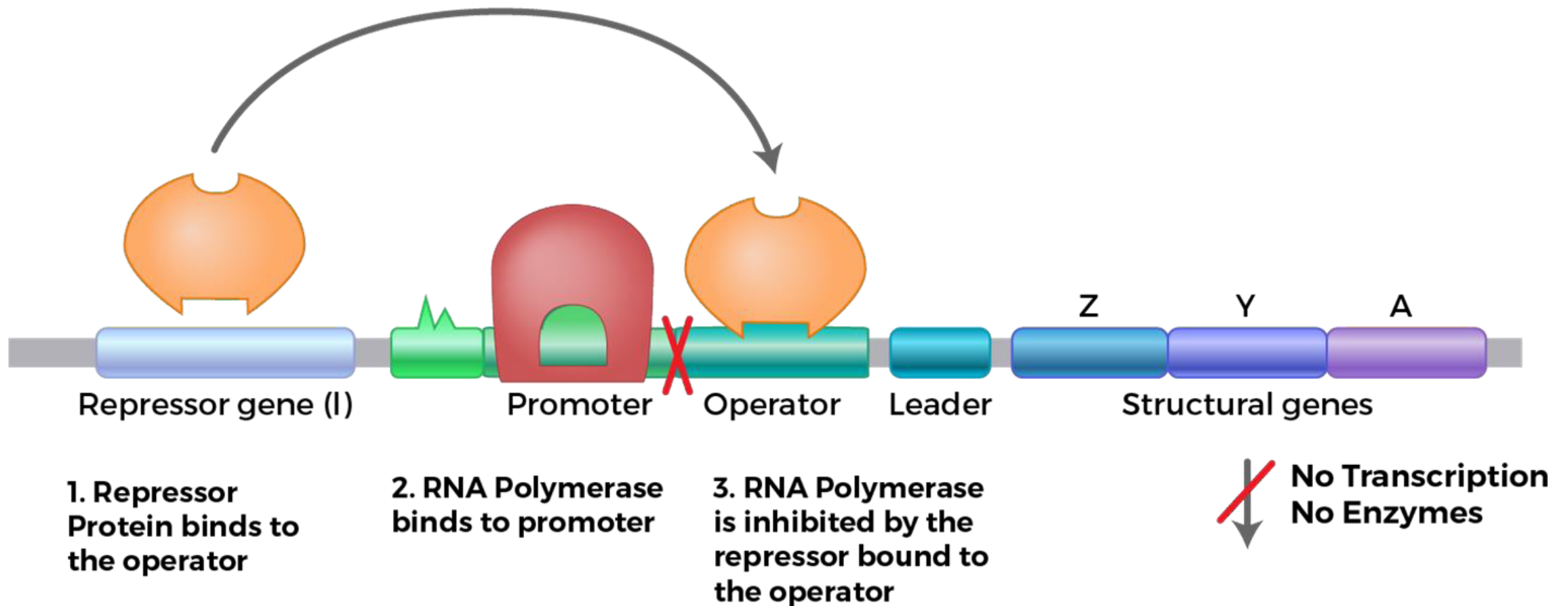


Control of the lac operon:

No lactose, repression

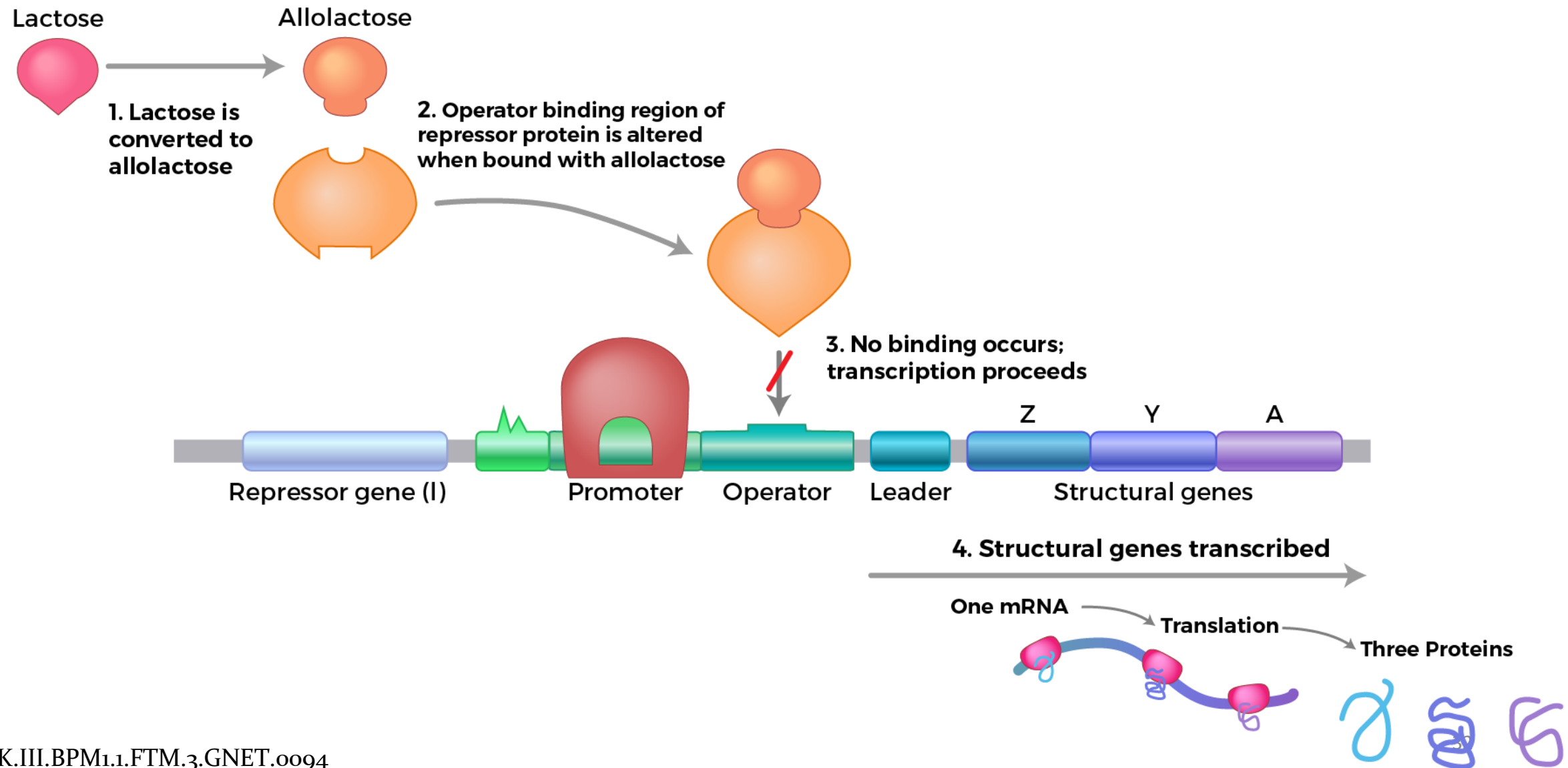
Lactose is present so you need the structural genes synthesized

Repression of the Lac Operon: No lactose present



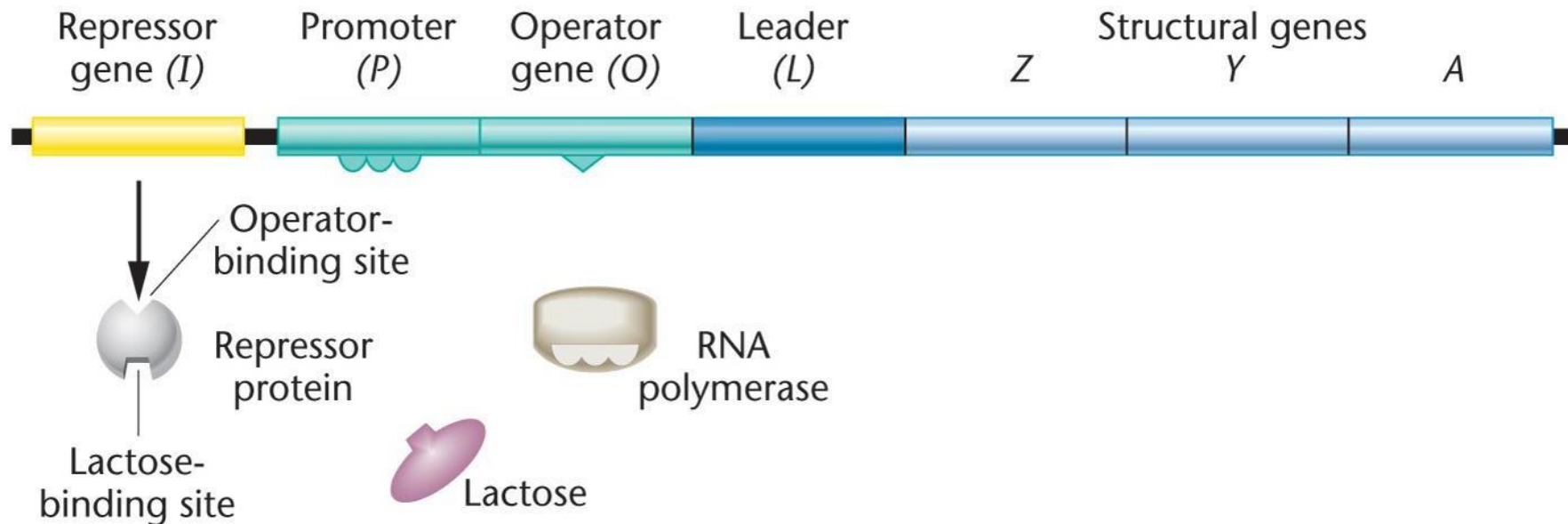
Absence of Lactose: Repressor Protein Binds to the Operator,
Blocking Transcription

Induction of the Lac Operon when lactose is present

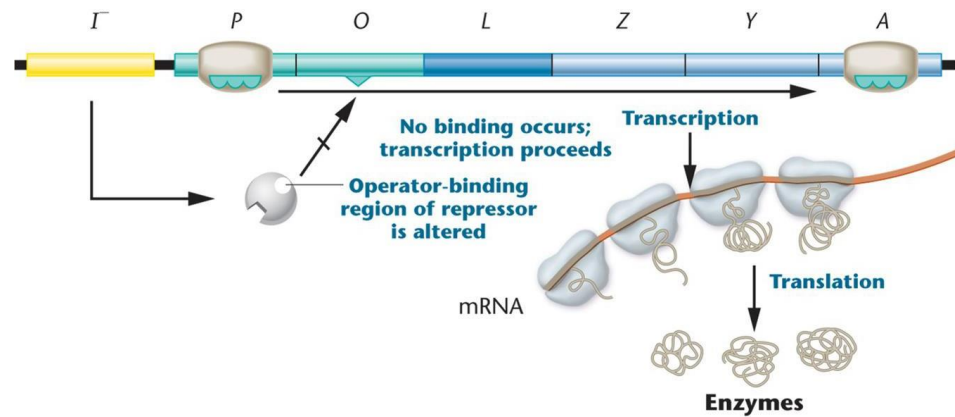


Jacob & Monod Mutational Analysis Summary

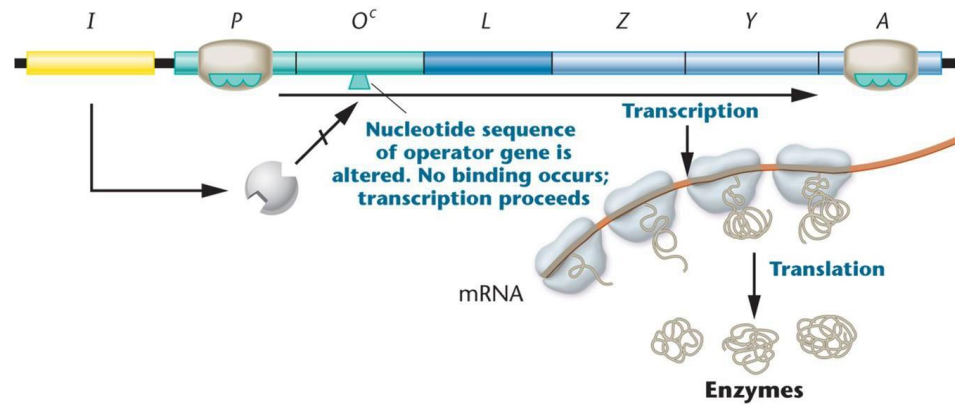
- ***lacP***⁻: Mutation in Promoter region and no structural enzymes synthesized
- ***lacZ***⁻, ***lacY***⁻ & ***lacA***⁻: Mutation in any of the structural genes and the protein is not made, lactose not metabolized



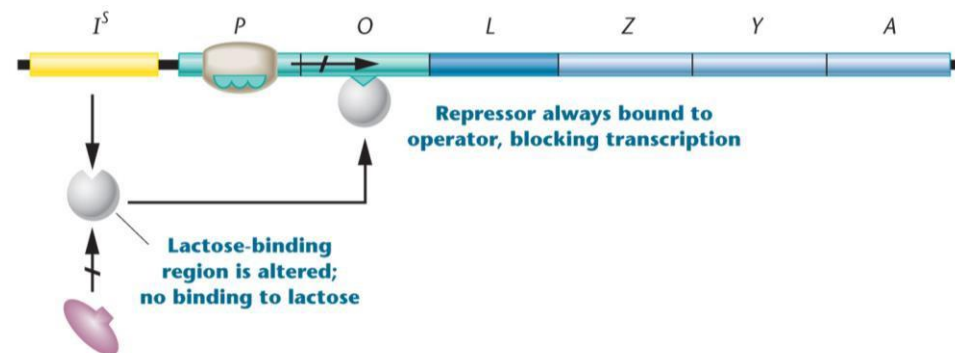
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- ***lacI*⁻**: non-functional repressor, unable to bind the operator to shut off transcription, this is a *constitutive* mutation



- ***lacO*^c**: Non-functional operator, repressor cannot bind. Since the system cannot be shut off, this is a *constitutive* mutation



- ***lacI*^s**: Super-repressor, unable to dissociate from operator. System always off

E. coli's Favorite Carbon Source is

Glucose

and will metabolize this sugar first even
in the presence of any other sugars

How does this work?

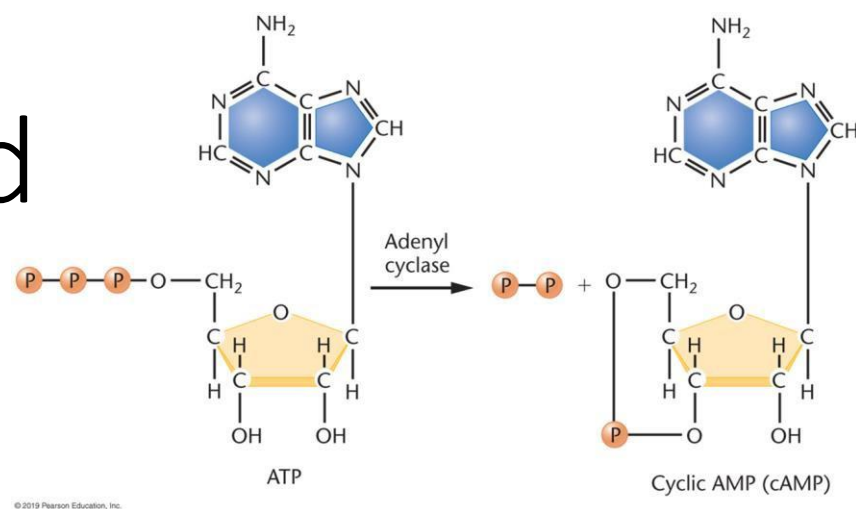
cAMP signaling

E. coli's Favorite Carbon Source is Glucose



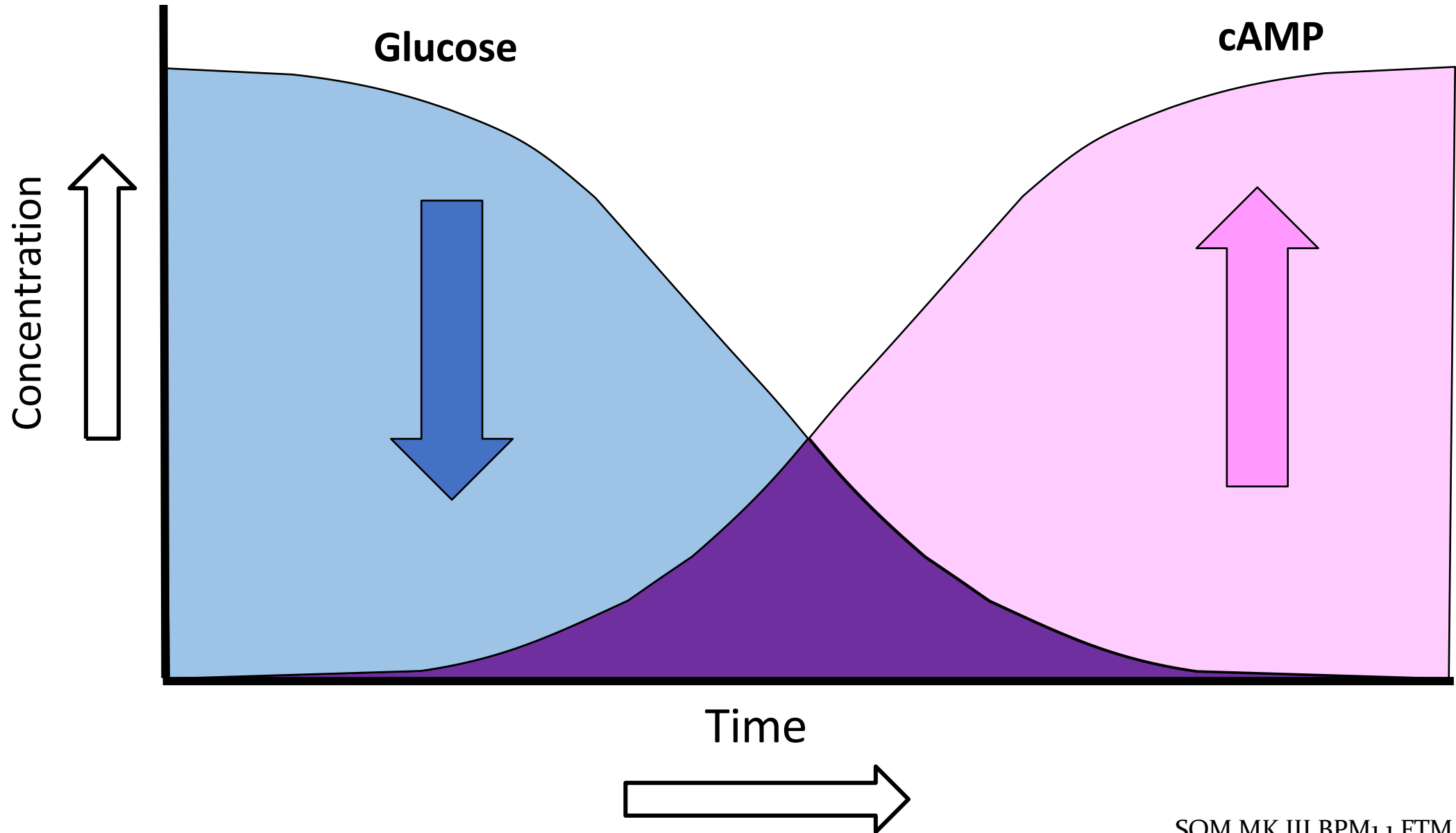
- While *E. coli* can metabolize lactose with the products of the *lac* operon, it actually 'prefers' glucose as a carbon source
- Therefore, when glucose is present, there is no need to express the genes in the *lac* operon
- **POSITIVE CONTROL**
 - Genetic expression of the *lac* operon occurs only if a regulator molecule directly stimulates RNA production (that molecule is cyclic AMP)

Glucose and cAMP

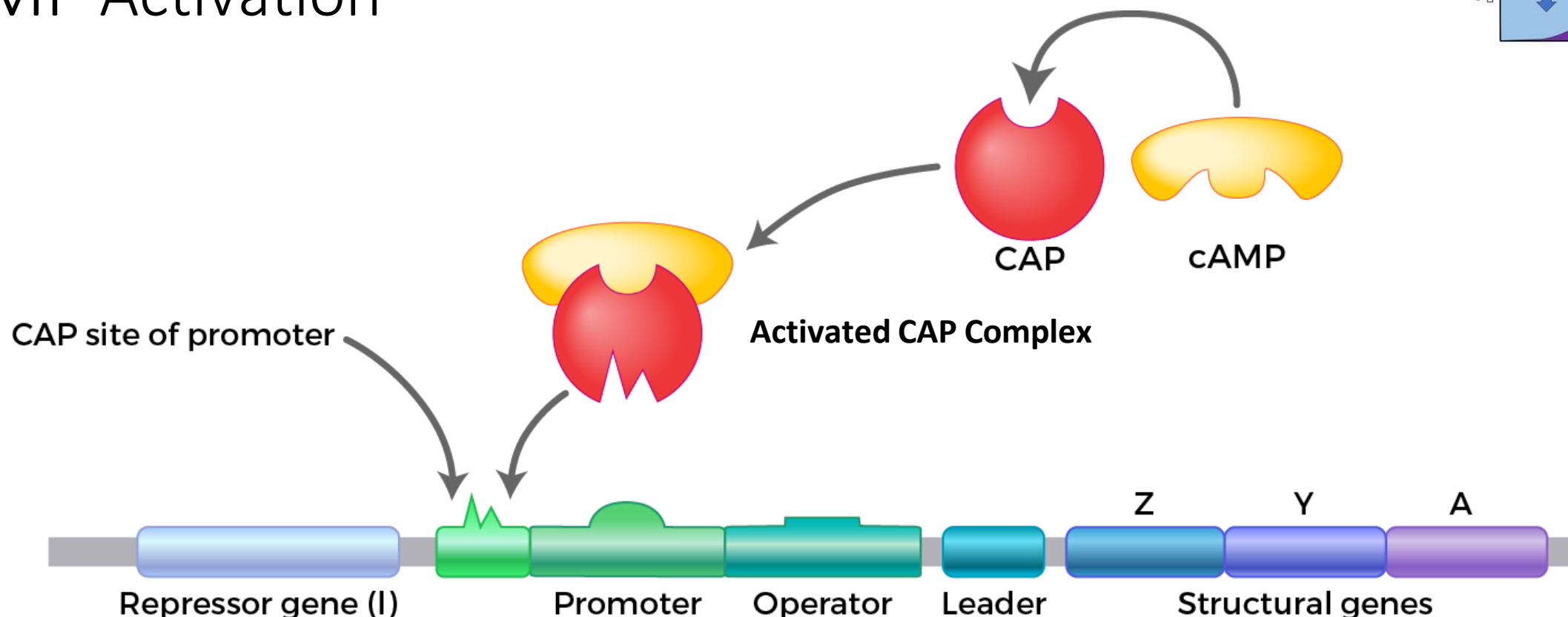
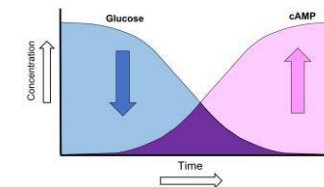


- If a cell runs out of glucose, a small molecule (cyclic AMP or cAMP for short) is produced by active **adenyl cyclase** (converts ATP to cAMP)
- cAMP is a 'hunger signal' that permits the expression of genes that break down other sugars, including lactose
- cAMP binds the activator protein **CRP** (cAMP receptor protein) or **CAP** (catabolite activator protein), which can then bind near *lacP* to **help activate transcription**

Glucose and cAMP



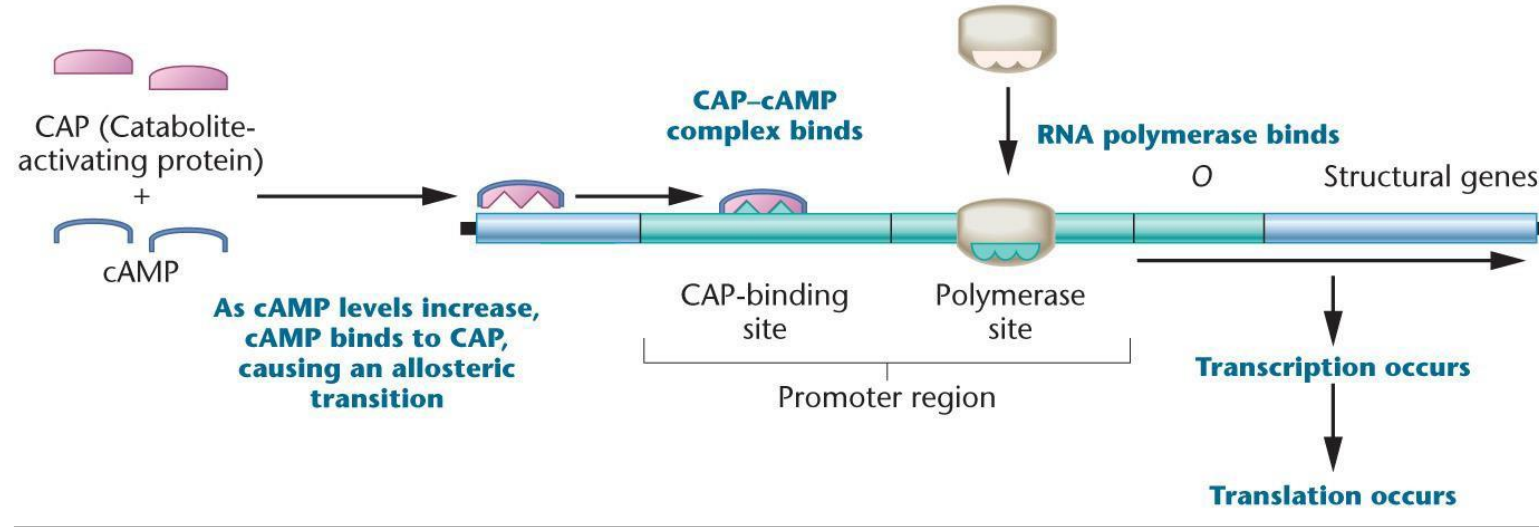
cAMP Activation



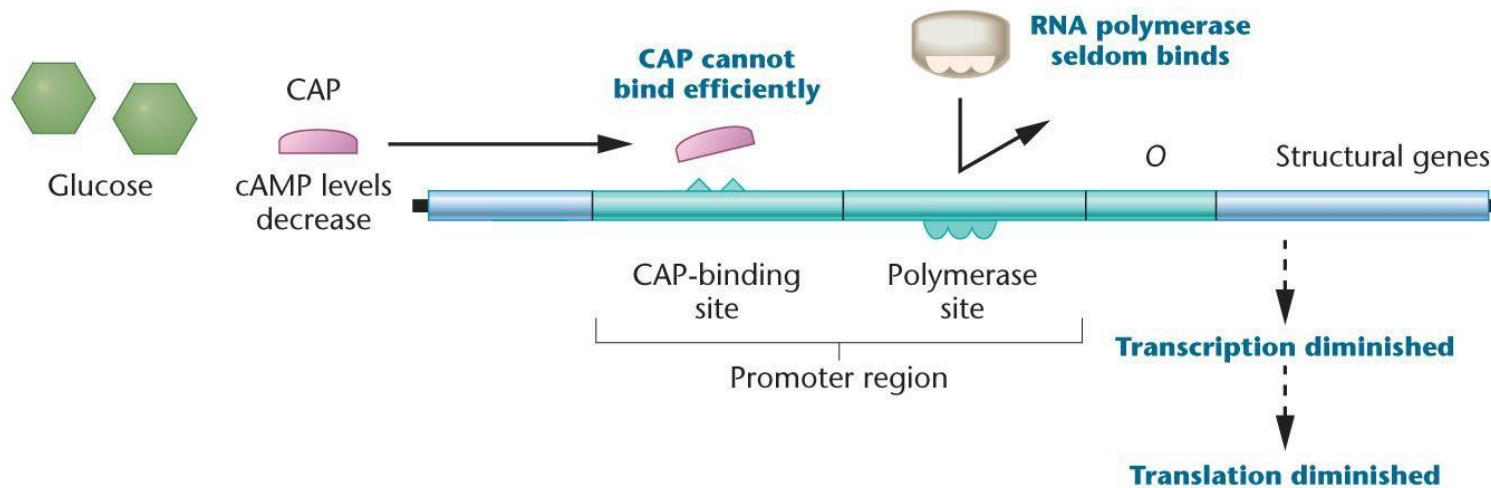
- In the absence of glucose, cAMP levels increase, resulting in the formation of cAMP-CAP complex
- cAMP/CAP complex binds to the CAP site of the promoter
- The binding of cAMP to the CAP protein inducing a conformational change to the complex such that it binds upstream of the promoter region of the lac operon
- High transcription of the structural genes is stimulated

Catabolite activation and repression Summary

(a) Glucose absent



(b) Glucose present



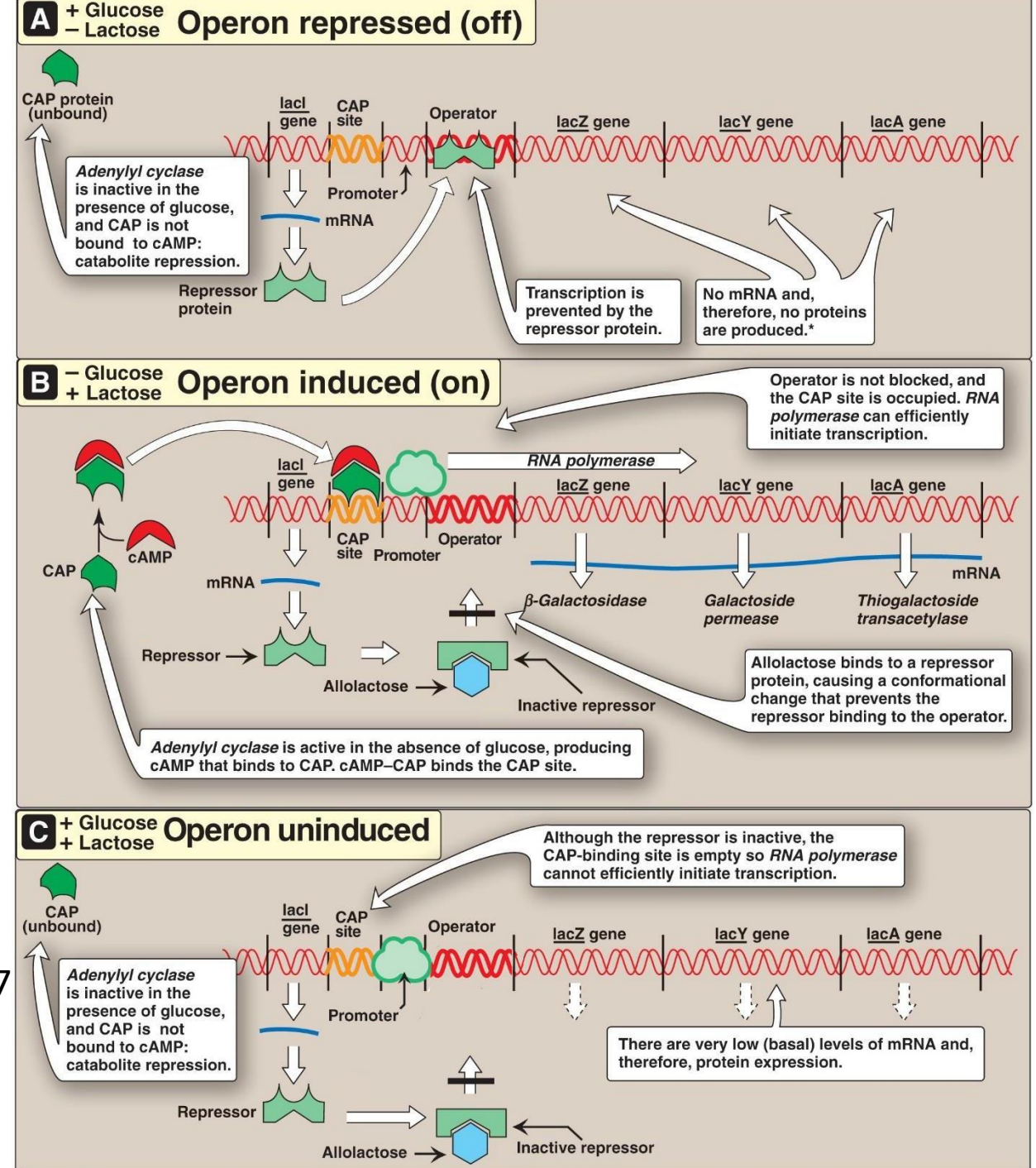
(a) In the absence of glucose, cAMP levels increase, resulting in the formation of a cAMP–CAP complex, which binds to the CAP site of the promoter, stimulating transcription.

(b) In the presence of glucose, cAMP levels decrease, cAMP–CAP complexes are not formed, and transcription is not stimulated.

Lac Operon Summary:

The *Lac* Operon will only be transcribed when lactose is present (repressor unbound the operator) and glucose is absent (CAP/cAMP is bound to CAP binding site).

Lippincott 8th ed. Page 517



On vs. Off

	Glucose present	Glucose absent
Lactose present	No repressor bound No activator bound OFF Some small basal transcription	No repressor bound Activator bound ON
Lactose absent	Repressor bound No activator bound OFF	Repressor bound Activator bound OFF

The end



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